

- 2 Scientists are researching new ways to reduce the global atmospheric carbon dioxide (CO₂) concentration. There are concerns that an increasing atmospheric CO₂ concentration may lead to effects that decrease biodiversity.

(a) Give **one** example of a human activity, other than deforestation, that contributes greatly to the increase in global atmospheric CO₂ concentration.

..... [1]

Algae are aquatic photosynthetic protocists. Some researchers genetically modified the unicellular alga, *Chlorella vulgaris*, to try to increase the rate of the light independent stage of photosynthesis.

C. vulgaris was modified to increase the expression of the gene coding for aldolase. Aldolase is an enzyme that causes an increase in the concentration of rubisco.

Two cultures of *C. vulgaris*, one that was not genetically modified (unmodified) and one genetically modified, were grown under controlled conditions for 14 days. Samples were taken from the cultures at regular intervals during the 14 days to obtain measurements of dry mass.

The results are shown in Fig. 2.1.

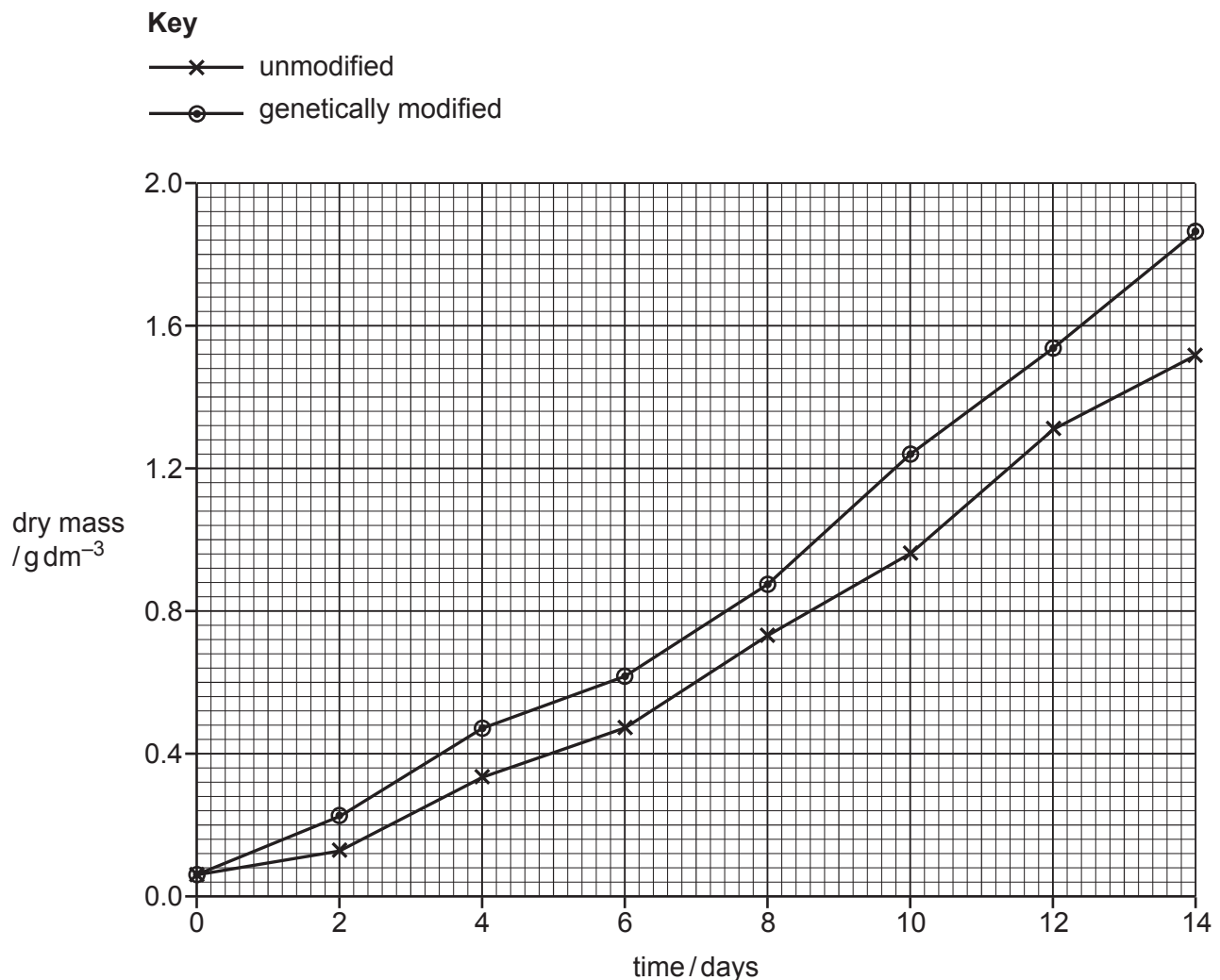


Fig. 2.1

- (b) With reference to Fig. 2.1, describe the differences between the results for the two cultures.

.....

.....

.....

.....

.....

..... [3]

- (c) Explain how the Calvin cycle was affected by the genetic modification of *C. vulgaris*.

.....

.....

.....

.....

.....

..... [3]

- (d) Intermediate products of the Calvin cycle are needed to produce organic molecules for use by the cell.

Describe how these organic molecules are used by cells.

.....

.....

.....

.....

.....

..... [3]

- (e) Planting large numbers of trees is one way to reduce global atmospheric CO₂ concentration. Large scale culture of genetically modified *C. vulgaris* could also reduce global atmospheric CO₂ concentration.

Suggest **one** advantage of using genetically modified *C. vulgaris* instead of trees to reduce global atmospheric CO₂ concentration.

.....

..... [1]

[Total: 11]